VF-B06 Split Tear

Test Method for Split Tear	Test method #: VF- B06 REF #: TM-R-041
Date Issued: November 18, 2022 Revision #: 4	Reason for Revision: Rectify thickness gauge capable of measuring to 0.01mm

PURPOSE AND SCOPE: To measure the load required to tear a material along its cross-section.

REFERENCES: ASTM D3574 SATRA TM65

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638
2. Clamping jaws, preferably rubber-faced
3. Foam
4. Vernier Caliper, capable of measuring to 0.01mm or thickness gauge, capable of measuring to 0.01mm.

SAMPLE PREPARATION:

1. Condition the test sample in a standard controlled environment @ 23.0± 2.0˚C temperature and at 55±10 % RH for at least 24 hours.
2. Die cut a test sample: width 25 mm, length 75 mm. The sample thickness must be 10.0 mm.
3. Draw a “tear line” lengthwise, at the midpoint of the thickness of the foam
4. Using a sharp blade or razor, hand cut a slit at each end of the test sample which is approximately 25 mm in length.

Note: When testing a midsole foam, refer to **Appendix C** for “Skin-on /Skin-off” designation.

REPETITIONS: Test three (3) samples per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Zero and calibrate the tensile testing machine.
2. Set the tensile testing machine to a crosshead speed of 50mm/minute, and a jaw gap of 4.0±0.5cm.
3. Fold back the two tabs at one end of the test sample, Insert and clamp one tab centrally in each of the top and bottom jaws, such that a minimal amount of tension is present on the tear line, and the sample points to the rear of the tensile testing machine.
4. Operate the tensile testing machine at a crosshead speed of 50mm/minute. If possible, allow the sample to tear with a distance 50mm but do not cut – allow it to tear freely.
5. Record the maximum load (L) reached during the test.
6. If there is premature breakage, do the test at the other end of the sample in the same manner described in **Step 3** and repeat **Steps4-6.**

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CALCULATION:

$$\textbf{Split tear strength (kgf/cm)} = \frac{L}{2.5\ cm}$$

REPORT: The average of the 3 samples to the nearest 0.1kgf/cm.